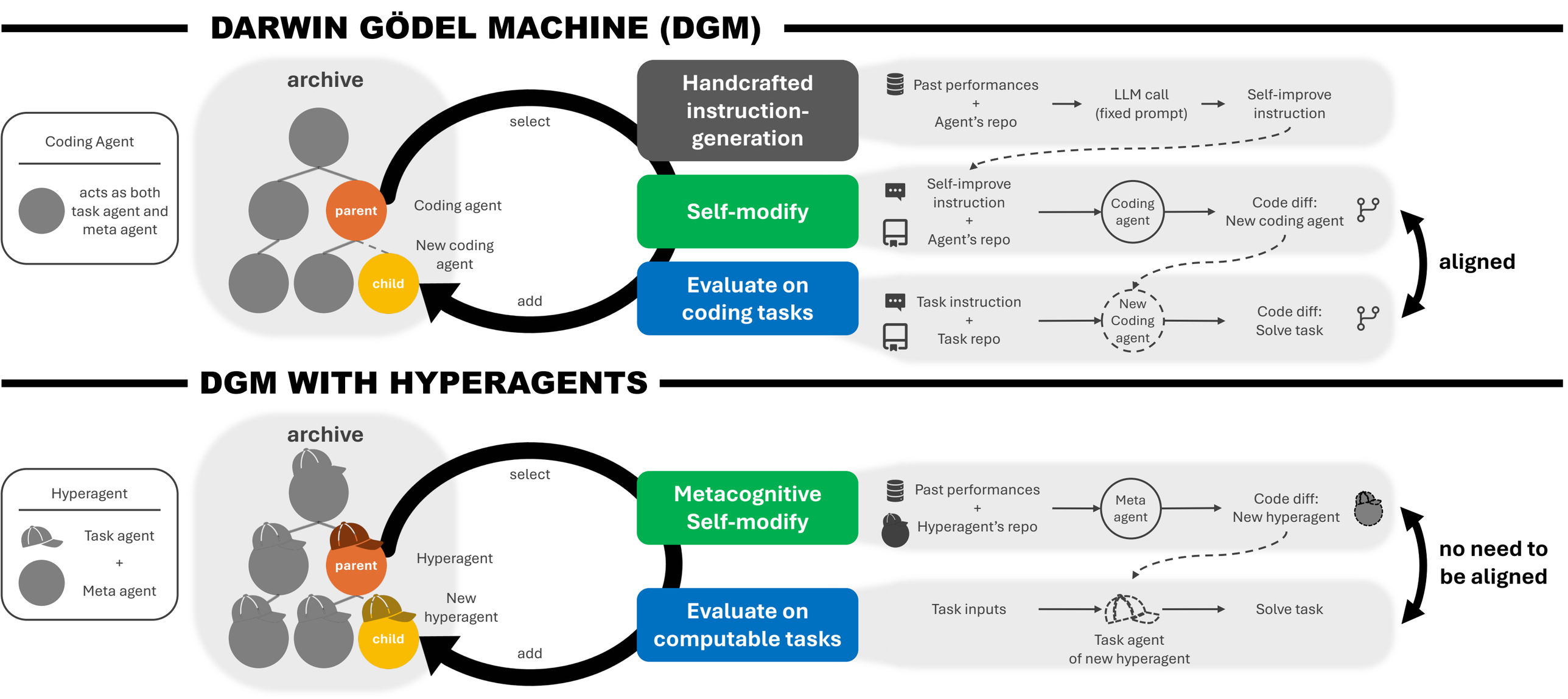
DARWIN GÖDEL MACHINE (DGM)
Coding Agent
acts as both task agent and meta agent
archive
parent
child
select
Coding agent
New coding agent
add
Handcrafted instruction-generation
Self-modify
Evaluate on coding tasks
Past performances + Agent's repo
LLM call (fixed prompt)
Self-improve instruction
Self-improve instruction + Agent's repo
Coding agent
Code diff: New coding agent
Task instruction + Task repo
New Coding agent
Code diff: Solve task
aligned
DGM WITH HYPERAGENTS
Hyperagent
Task agent + Meta agent
archive
parent
child
select
Hyperagent
New hyperagent
add
Metacognitive Self-modify
Evaluate on computable tasks
Past performances + Hyperagent's repo
Meta agent
Code diff: New hyperagent
Task inputs
Task agent of new hyperagent
Solve task
no need to be aligned